

DENİZ EFE KARACAKAYA

Antalya, Türkiye | +90 530 861 75 14 | denizefekarakakaya14@gmail.com
github.com/Denizfe | linkedin.com/in/deniz-efe-karakakaya-44858531b

SUMMARY

Artificial Intelligence & Data Engineering undergraduate and AI/MLOps engineering intern with hands-on experience building cloud-native, AI-powered, and DevOps-driven systems. Co-developed a privacy-preserving fraud detection research program using federated learning (Flower, PyTorch, SHAP) and a self-hosted Kubernetes/GitOps MLOps platform. Skilled in Python, Java, TypeScript, and production-style projects spanning e-commerce, generative AI, and MLOps infrastructure. Seeking Software Engineering, AI/ML, or MLOps roles.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, Bash, YAML

AI & Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, Federated Learning (Flower), MLflow, Pandas, NumPy, OpenCV, Prophet, Google Gemini API, Google AI Studio

Frontend: React, Next.js, Angular, Tailwind CSS, shadcn/ui

Backend: Spring Boot, Node.js, Express, FastAPI, Flask, REST APIs, gRPC, JWT Authentication

Cloud & DevOps: Docker, Kubernetes, Helm, Argo CD, GitOps, GitHub Actions, Actions Runner Controller (ARC), GHCR, CI/CD, Sealed Secrets, Google Cloud, AWS

Databases: PostgreSQL, MinIO, Hibernate/JPA, MongoDB, Supabase, Firebase

Tools: Git, Linux, VS Code, Figma

EXPERIENCE

AI & MLOps Engineering Intern — ASEE Information Technologies

Jun 2026 – Present

Istanbul, Türkiye

- Co-developed a privacy-preserving fraud detection research PoC on the IEEE-CIS transaction dataset (590K rows), benchmarking six models (XGBoost, CatBoost, Random Forest, MLP, LightGBM, Logistic Regression); XGBoost achieved the top F1 score (0.494).
- Applied SHAP-based feature selection to cut the feature schema by 95% (391 → 20 columns) while retaining 98.2% of baseline F1, enabling a minimal cross-bank data-visibility contract.
- Built a cross-institution Federated Learning pipeline with Flower (FedAvg) and a PyTorch MLP over real gRPC; ran a 47-configuration benchmark across aggregation strategies (FedProx, FedAdam, FedYogi, SCAFFOLD), data splits, and personalization methods (Ditto, FedPer).
- Identified a fraud-prevalence-weighted FedAvg configuration matching the strongest individual bank's F1 while lifting the weakest bank's F1 by +0.17, and extended validation to the AMLWorld dataset with a Graph Neural Network variant.
- Designed and deployed a self-hosted MLOps platform on Kubernetes using GitOps (Argo CD, Helm), with an MLflow tracking server (PostgreSQL + MinIO) and CI/CD pipelines automated via GitHub Actions/ARC, GHCR, and Sealed Secrets.

PROJECTS

Chronos — Luxury Watch E-Commerce Platform

Full-stack academic project

- Architected a secure, role-based e-commerce platform on Google Cloud using Angular 17, Spring Boot, and PostgreSQL, with JWT-based authentication.
- Integrated Stripe payment processing and a Flask/Gemini AI chatbot hardened with prompt-injection safeguards.
- Designed a normalized (3NF) relational database and built a Python ETL pipeline to load and manage 1,500+ product records.

Technologies: Java, Spring Boot, Angular 17, JWT, Hibernate, PostgreSQL, Google Cloud SQL, Stripe, Flask, Gemini API

Local MLOps Platform — GitOps on Bare Metal

- Personal project extending the ASEE MLOps stack: self-hosted K3s cluster with GitOps-driven deployments.

Technologies: Docker, Kubernetes (K3s), Argo CD, Helm, GitHub Actions, MinIO, Sealed Secrets, GitOps

Gym Buddy AI — Voice-Driven Fitness Coach

- Built an AI-powered fitness and nutrition assistant that delivers real-time voice guidance during workouts.

Technologies: React, TypeScript, Tailwind CSS, Google AI Studio, Supabase

Artifée — Generative AI Photo Editor

- Developed a generative photo-editing web app supporting object removal and image enhancement, powered by the Gemini API.

Technologies: Next.js, TypeScript, Tailwind CSS, Genkit, Gemini API

EDUCATION

B.S., Artificial Intelligence & Data Engineering — Akdeniz University

2024 – 2028 (Expected)

Antalya, Türkiye

- Coursework and applied projects in machine learning, object-oriented programming, data visualization, and full-stack development (Python, Java, Spring Boot, Angular).

CERTIFICATIONS

- Data Analysis School (Veri Analizi Okulu) — Artificial Intelligence & Machine Learning and Artificial Intelligence & Facilitator Tools modules; Turkish Higher Education Council (YÖK) / Marmara University, with METU, ITU & Boğaziçi University (2025–2026)
- BTK Academy / Information Technologies and Communication Authority — Artificial Intelligence (07/2025); Machine Learning Applications with Python (06/2025); Python Programming Language (06/2025)

LANGUAGES

Turkish: Native | English: Professional Working Proficiency | German: Basic | Spanish: Basic